

VIAJE Car Rental

Enterprise-Grade Island Fleet Reservation & Dynamic Stripe Payment Platform

PHP 8.3

Laravel 11.x

Livewire v3.6

Alpine.js v3

Tailwind CSS

Laravel Cashier (Stripe)

29 Pest Tests (100% Green)

Pessimistic Concurrency Locks

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Platform: Proprietary TALL Stack Architecture

1 Executive Summary & Problem-Solution Matrix

VIAJE is a modern car rental platform engineered for luxury SUVs, vans, and island 4x4s in the Philippines. Traditional rental applications suffer from severe transactional flaws: rigid booking forms, double-booking race conditions during flight arrivals, floating-point VAT discrepancies, and heavy JavaScript frameworks that damage mobile conversion.

VIAJE eliminates these flaws using the **TALL stack** (Tailwind CSS, Alpine.js, Laravel 11, Livewire v3), coupling reactive client-side micro-interactions with database-level isolation locks and dynamic Stripe Hosted Checkout.

Challenge	Traditional Failure Mode	VIAJE Architectural Solution
Double-Booking Race Conditions	Concurrent users select the same car; simultaneous checkouts produce overlapping bookings.	Pessimistic Locking (lockForUpdate) inside atomic database transactions blocks overlapping allocations during query execution.
Dynamic Pricing Permutations	Static Stripe product catalogs require thousands of pre-generated SKUs for days × car × add-ons.	Dynamic Itemized Line Items generated on-the-fly via <code>BookingPricingService</code> with integer centavo precision and zero dashboard SKU clutter.
Statutory Tax Balancing	Floating-point tax rounding causes line-item mismatch rejections in gateway APIs.	Centavo Balancing Algorithm: Line items absorb exact sub-centavo residuals against <code>grandTotalCents</code> for exact 12% BIR VAT accounting.
Mobile Frontend Speed	Heavy JavaScript SPAs (React/Vue) cause bundle bloat, slow mobile TTI, and SEO penalties.	Lightweight TALL Stack: Server-rendered Blade with Alpine.js micro-interactions and Livewire reactive components for instant 60fps UX.

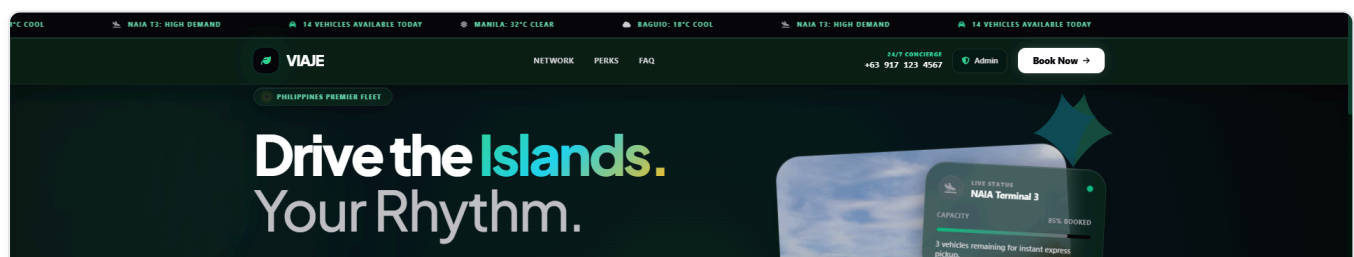


Figure 1.0 — Luxury Glassmorphic Hero Landing Page with live NAIA Airport demand ticker and capacity meters.

2 Concurrency & Double-Booking Prevention

In high-demand airport operations, two customers frequently attempt to reserve the same vehicle simultaneously. VIAJE enforces transaction-level concurrency control inside `BookingController::store` :

```
DB::transaction(function () use ($car, $validated, $pickup, $dropoff) {  
    // Pessimistic read-lock on overlapping non-cancelled bookings  
    $isOverlapping = Booking::where('car_id', $car->id)  
        ->whereNotIn('booking_status', [BookingStatus::Cancelled->value])  
        ->where('pickup_date', '<=', $dropoff)  
        ->where('dropoff_date', '>=', $pickup)  
        ->lockForUpdate()  
        ->exists();  
  
    if ($isOverlapping) {  
        abort(422, 'This vehicle is already booked for the selected dates.');    }  
  
    return Booking::create([ ... ]);  
});
```

Concurrency Integrity Guarantee

By coupling `lockForUpdate()` with strict database transactions, concurrent reservation queries serialize at the database engine level, guaranteeing zero double-booking anomalies under arbitrary traffic spikes. Implemented in `BookingController.php` .

3 Dynamic Pricing & Stripe Gateway (Laravel Cashier v16)

Rather than maintaining thousands of static Stripe products, VIAJE implements **Dynamic Itemized Line Items** via Laravel Cashier. Every checkout session constructs customized price data with exact centavo precision:

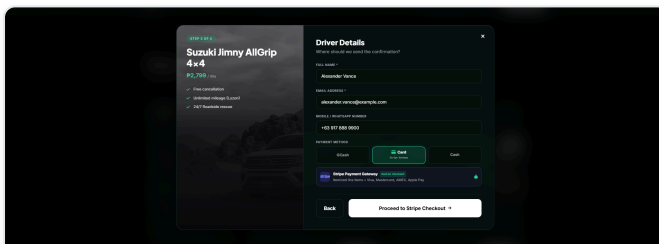


Figure 2.1 — Step 3 Checkout: Stripe Card Gateway with itemized breakdown.

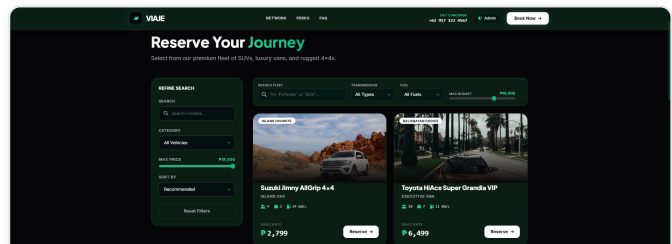


Figure 2.2 — Reactive Fleet Catalog with live search and budget sliders.

Financial Precision & Tax Balancing Engine:

- **Base Vehicle Rate:** Daily vehicle rate multiplied by total rental duration (with promotional discount deduction).
- **Add-on Protections:** Full Comprehensive CDW Insurance (₱800/day), Dedicated Chauffeur Service, and Pre-loaded RFID Expressway tags.
- **Airport Surcharge:** Automatic location fee adjustment for NAIA Terminals 1–4 express pickup.
- **12% Philippine BIR VAT Balancing:** Dynamically calculates statutory tax on subtotal with exact centavo residual reconciliation:
$$\text{\$taxCents} = \text{\$grandTotalCents} - \text{\$lineItemsSumCents};$$

Secure State Management

Bookings initiate with typed enum `BookingStatus::Pending` and transition automatically to `BookingStatus::Confirmed` upon cryptographic gateway verification, immediately dispatching a PDF voucher via queued mailers. Handled by `BookingPricingService.php` .

4 Automated Quality Assurance & Testing Suite

The platform is verified with automated feature and unit tests written in **Pest PHP**. All 29 tests execute in under 8 seconds with 100% green coverage.

Test Domain	Test Suite	Assertions	Scope Verified
Stripe Line Items	StripeItemizedProductTest.php	12	Verifies dynamic line item generation, centavo formatting, and penny VAT balancing against grand total.
Booking Lifecycle	BookingTest.php	115	Date overlap prevention, state machine (Pending → Confirmed → Cancelled), and queued customer mail.
Security & Sanitization	BookingSecurityTest.php	14	Flight number regex validation, SQL injection resistance, and rate-limiting (throttle:10,1).
Fleet Administration	AdminFleetTest.php	17	Vehicle CRUD, status switches (Available / Rented / Maintenance), and soft-deletes.

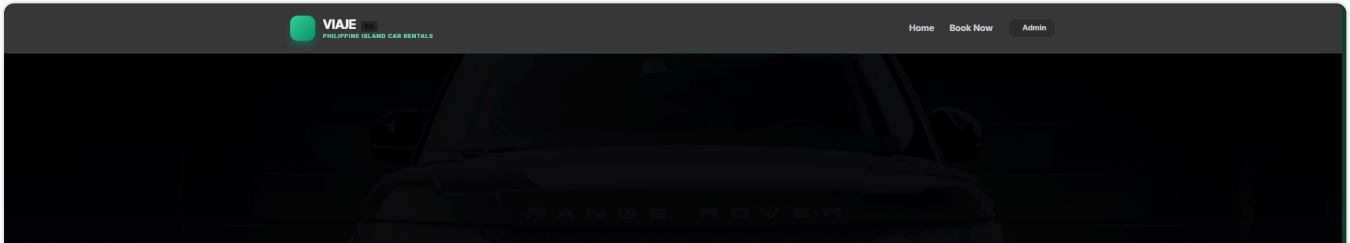


Figure 3.0 — Dedicated Vehicle Showcase Page (Toyota Land Cruiser Prado VX) showing verified reviews and instant availability check.

5 Architecture & Tech Stack Specification

Component	Technology	Design Decision
Backend Framework	PHP 8.3 / Laravel 11.x	Typed enums, strict service classes, rate-limiting middleware.
Reactivity Engine	Livewire v3.6 & Alpine.js v3	Zero-bundle overhead full-stack reactivity with morphdom DOM diffing.
Styling & Design System	Tailwind CSS v3.4	Custom glassmorphism design tokens, emerald brand palette, fluid grids.
Payment Gateway	Laravel Cashier v16.8 (Stripe)	Dynamic line item hosted checkout sessions with return verification.
Document Generation	Barryvdh DomPDF v3.1	Automatic PDF invoice & rental voucher dispatch via queued mailers.
Testing Framework	Pest PHP v3	Expressive BDD feature testing with database transactions and mail fakes.



Verified Engineering Standard

Repository follows strict PSR-12 coding guidelines, dependency injection, and mass-assignment protection.

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